ARGOS H1 engine ECU v2.1 (Teensy 4.1, separate from the flight computer, CAN 1 Mbit/s)

Six hierarchical sheets; nets connect by global labels and power symbols (net names = interface names in REPORT.md).
Generated 2026-10-12 by hardware/kicad_gen/make_ecu_kicad.py from hardware/kicad_gen/ecu_design.py, then editable by hand.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4). Files use the KiCad 7 format and open in KiCad 7, 8 and 9 (8/9 upgrade on save).
ERC: KiCad 7 eeschema ERC run headless (kicad-cli 7 has no sch erc); report: erc/argos_ecu_erc.rpt.
All symbols live in the project library argos_ecu.kicad_sym (flattened copies of KiCad 7.0.11 stock symbols + custom).
Status: design schematic only. Not built, no PCB layout routed, no bench test. ERC clean is not hardware validation.
Safety chain: arming plug (physical) -> fused ARM bus switch = ARM_REQ AND watchdog OK AND OVP_OK -> ARM-only 5 V gate rail.
OVP_OK: LMV331 comparator on PC_SIG (~31 bar g, ratiometric) clears a 74LVC1G175 latch; only an ARM_REQ rising edge sets it.

Power

File: power.kicad_sch

MCU_CAN

File: mcu_can.kicad_sch

Safety_Arming

File: safety_arming.kicad_sch

Actuators

File: actuators.kicad_sch

Igniter_Valves

File: igniter_valves.kicad_sch

Sensors

File: sensors.kicad_sch

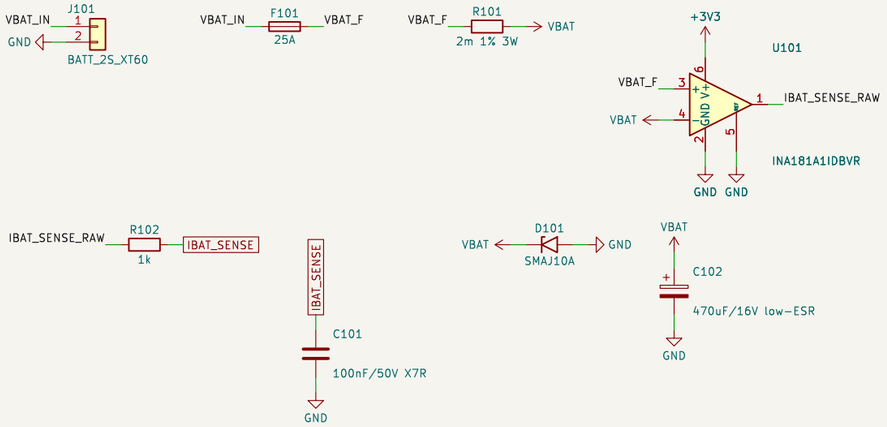
Status: design schematic, not hardware-validated
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Root: overview and sheet index
ARGOS H1 engine - 06-electronics

Sheet: /
File: argos_ecu.kicad_sch

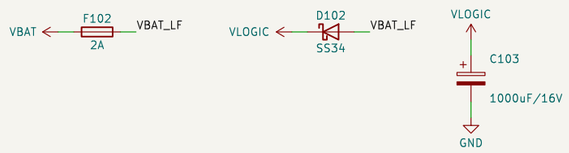
Title: ARGOS engine ECU v2.1 - root

Size: A3	Date: 2026-10-12	**Rev: v2.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/7

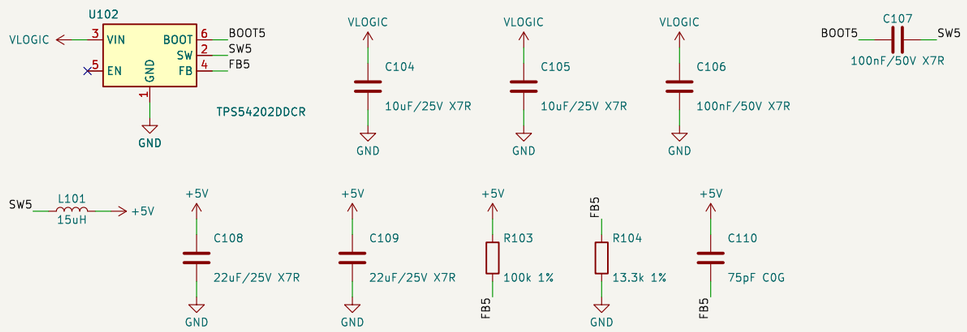
Battery input, main fuse, current sense, transient clamp



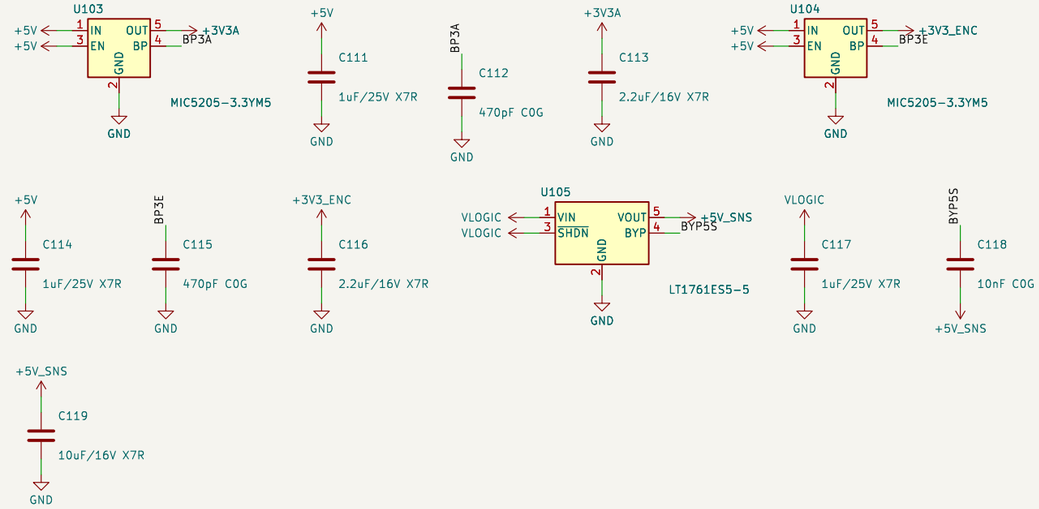
Logic supply path: 2 A fuse, Schottky, 1 mF hold-up



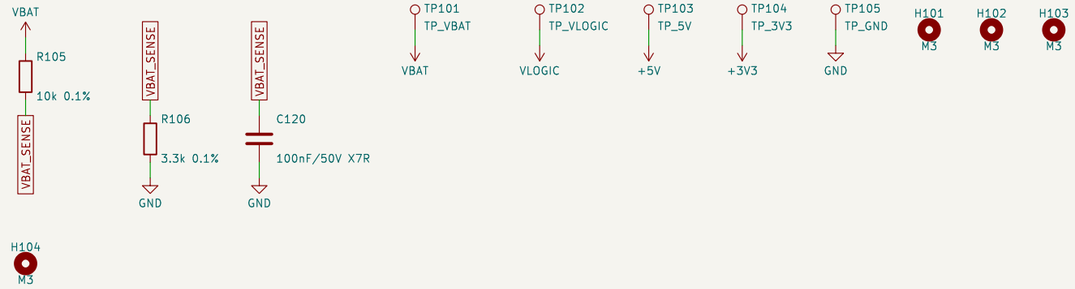
5 V buck TPS54202 (TI SLVSD26C 5 V design: 15 uH, 2x22 uF, 100k/13.3k, 75 pF)

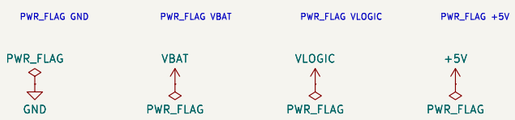


Low-noise LDOs: +3V3A (ADC AVDD), +3V3_ENC (encoder cables), +5V_SNS (transducers)



Battery voltage sense, test points, mounting





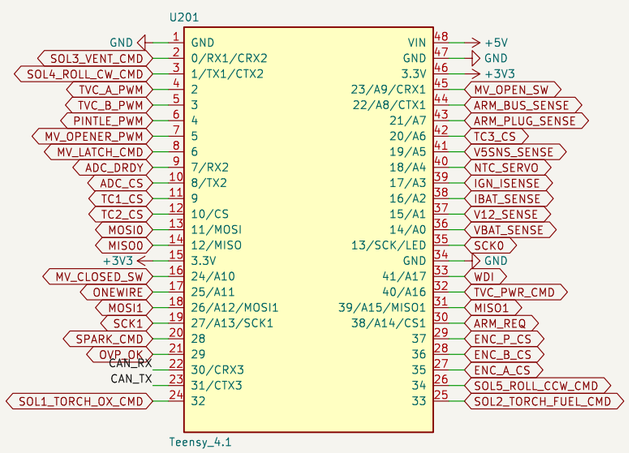
Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-12); edit freely in KiCad
ARGOS H1 engine — 06-electronics

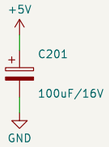
Sheet: /Power/
File: power.kicad_sch

Title: ARGOS ECU v2.1 — Power input, protection, rails

Size: A3	Date: 2026-10-12	**Rev: v2.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/7

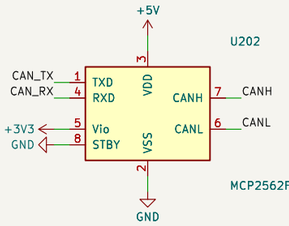
MCU: PJRC Teensy 4.1 on 2x24 sockets (cut VUSB–VIN pads; USB is service only)

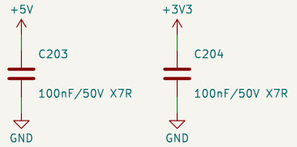




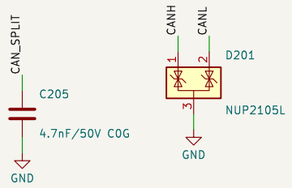


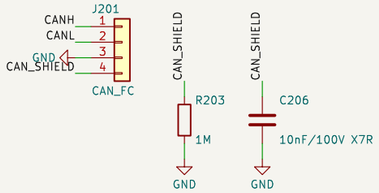
CAN FD transceiver MCP2562FD (CAN3, 1 Mbit/s classic, FD-ready), split termination, ESD











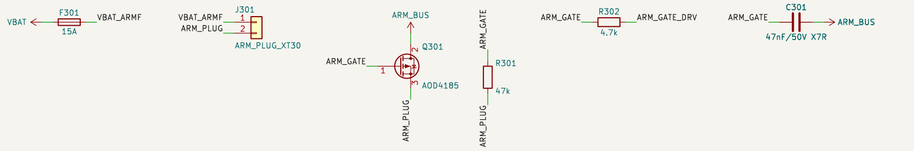
Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-12); edit freely in KiCad
ARGOS H1 engine — 06-electronics

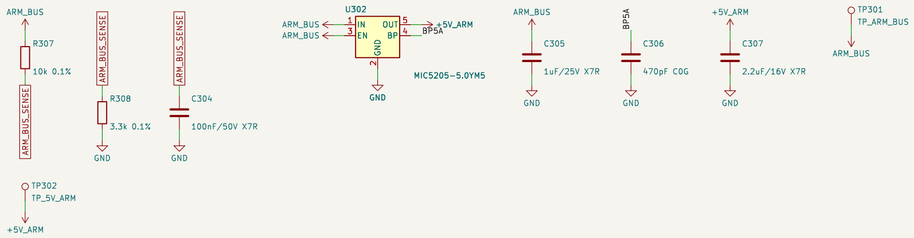
Sheet: /MCU_CAN/
File: mcu_can.kicad_sch

Title: ARGOS ECU v2.1 — MCU Teensy 4.1 and CAN

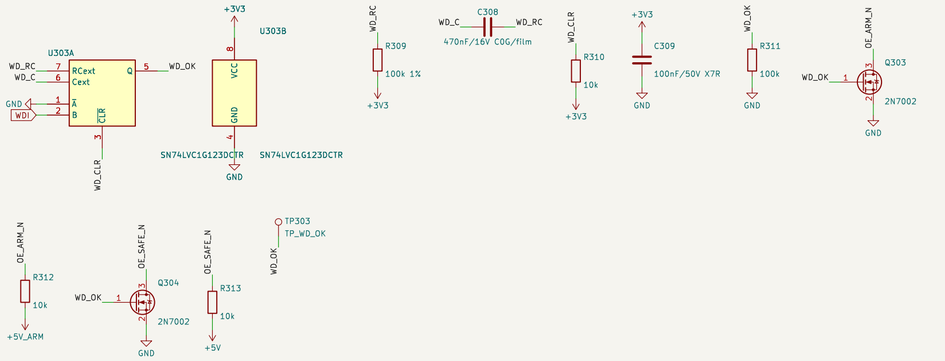
Size: A3	Date: 2026-10-12	**Rev: v2.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/7

Arming plug (physical break) and ARM high-side switch (soft start)

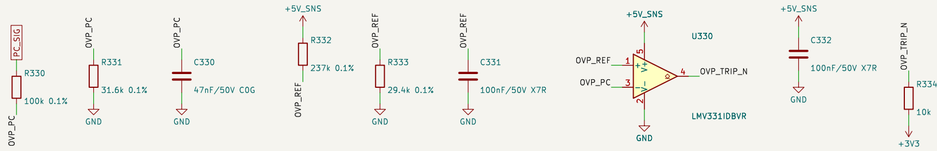


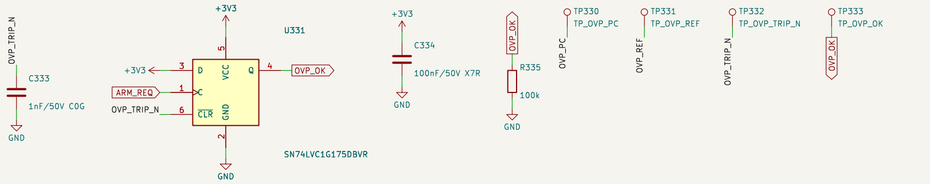


Independent hardware watchdog SN74LVC1G123 (tw = K R C = 1.0–1.1 x 100k x 470nF = 47–52 ms)

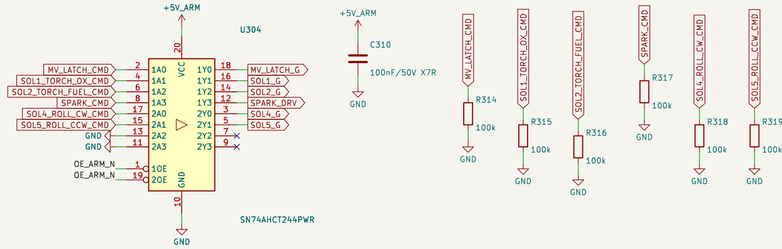


Hardware chamber over-pressure trip (~31 bar g): LMV331 -> SN74LVC1G175 latch -> ARM AND; re-armed only by an ARM_REQ rising edge

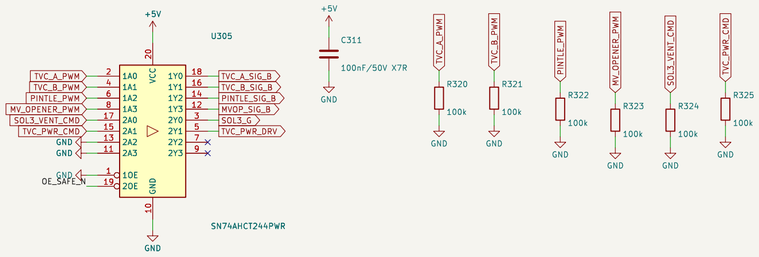




ARM-gated gate buffer U304 SN74AHCT244 on +5V_ARM (/OE = watchdog)



SAFE-allowed buffer U305 SN74AHCT244 on +5V: servo PWM 3.3->5 V (always on), vent + TVC power (watchdog)





Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-12); edit freely in KiCad

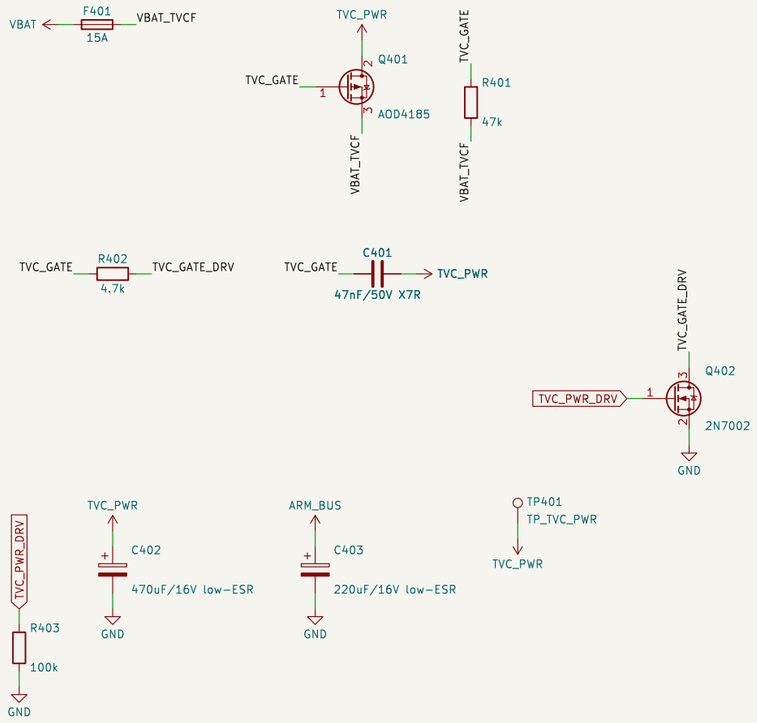
ARGOS H1 engine — 06-electronics

Sheet: /Safety_Arming/
File: safety_arming.kicad_sch

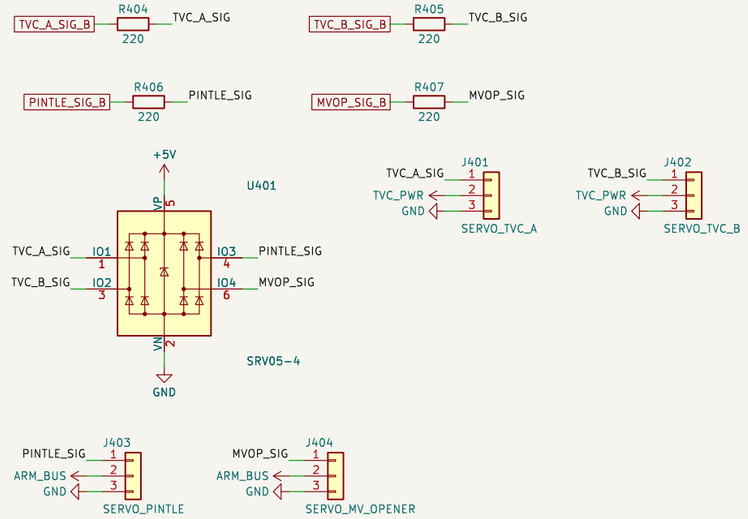
Title: ARGOS ECU v2.1 — Arming, watchdog, output buffers

Size: A2	Date: 2026-10-12	**Rev: v2.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7

TVC servo power switch (watchdog-gated, soft start) and reservoirs



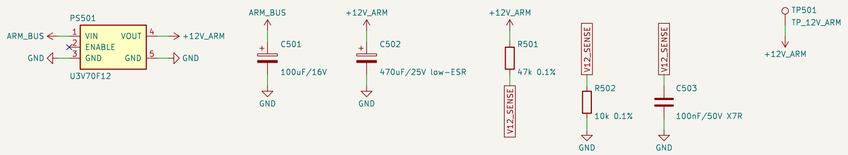
Servo connectors (Micro-Fit 3.0: 1 signal, 2 V+, 3 GND), 220 ohm series, ESD



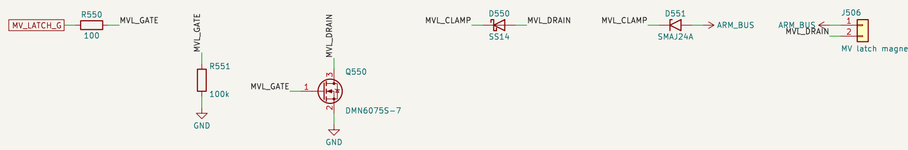
Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-12); edit freely in KiCad
ARGOS H1 engine — 06-electronics

Sheet: /Actuators/		
File: actuators.kicad_sch		
Title: ARGOS ECU v2.1 — Servo actuators, TVC power switch		
Size: A3	Date: 2026-10-12	**Rev: v2.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/7

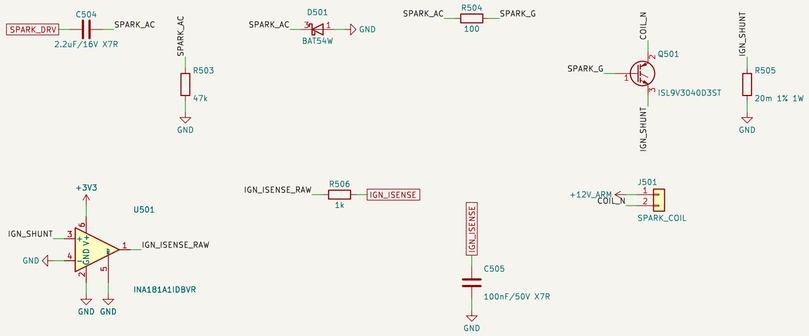
12 V boost on ARM bus (Pololu U3V70F12): spark coil, torch and roll solenoids



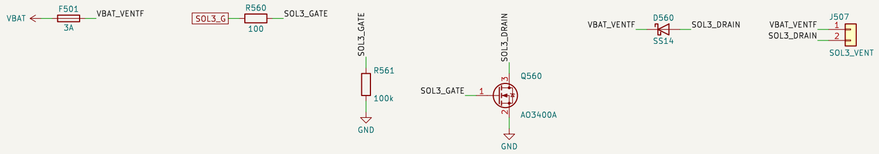
Main-valve holding magnet (ARM bus, fast release)



Spark coil driver: logic-level ignition IGBT, AC-coupled gate (hardware max dwell), current sense



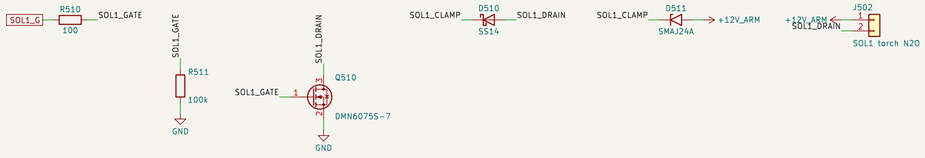
Tank vent solenoid (VBAT, PWM-capable, SAFE-allowed, watchdog-gated)

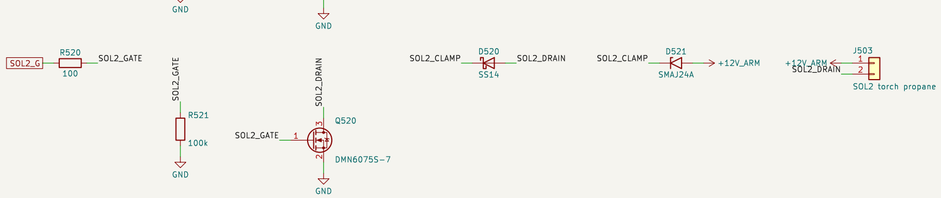


Main-valve position switches (to GND when actuated)

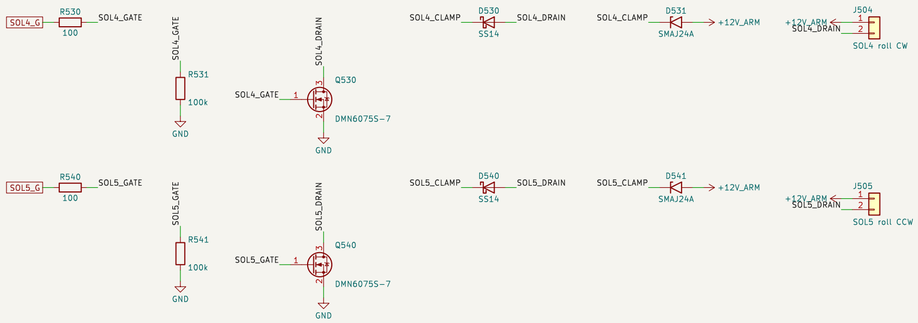


Torch solenoids (12 V, ARM-gated)





Roll-control solenoids (GNC v1 provision, 12 V, ARM-gated)



Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-12); edit freely in KiCad

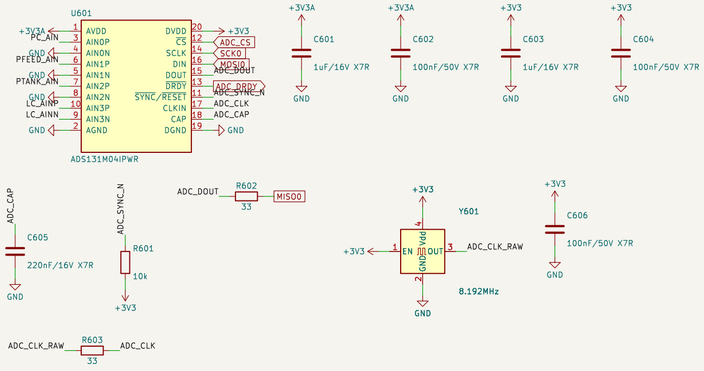
ARGOS H1 engine — 06-electronics

Sheet: /Igniter_Valves/
File: igniter_valves.kicad_sch

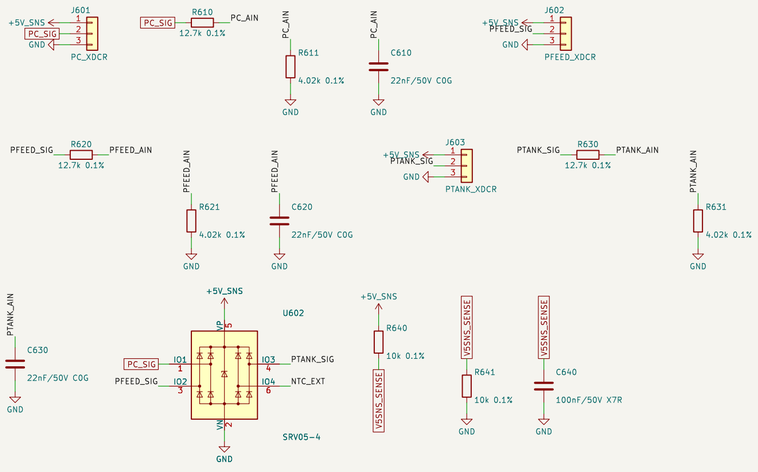
Title: ARGOS ECU v2.1 — 12 V boost, spark, solenoid drivers

Size: A2	Date: 2026-10-12	**Rev: v2.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/7

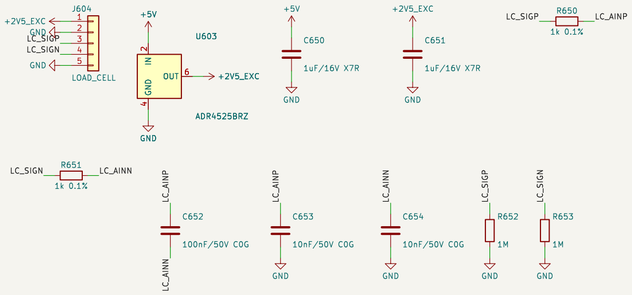
24-bit ADC ADS131M04 (4 kSPS, decimated to 1 kHz in firmware), 8.192 MHz clock



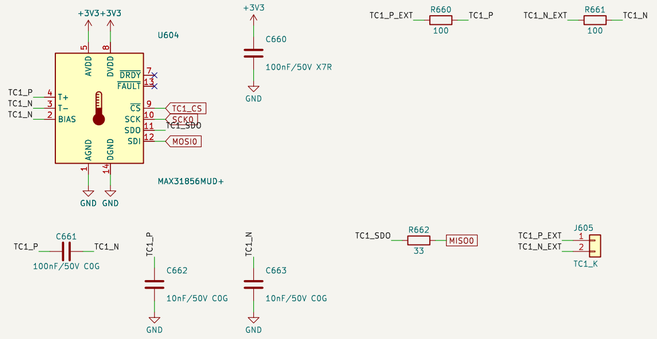
Pressure transducers (0.5–4.5 V ratiometric on +5V_SNS): divider + anti-alias RC



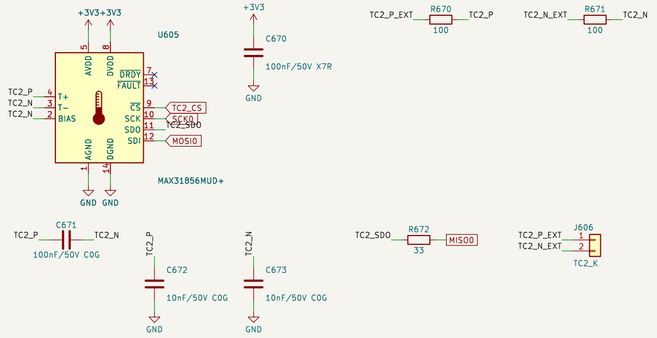
Load cell (test stand, ADC CH3 PGA 64–128) with 2.5 V precision excitation



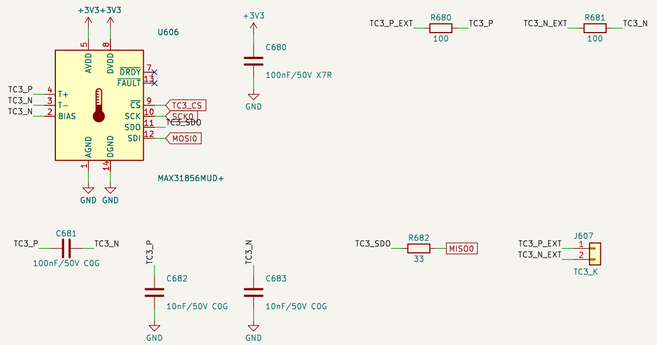
Thermocouple 1 (case forward): MAX31856, K type, 100 ohm + 100 nF + 10 nF filter (Maxim 19-7534 Fig. 8)



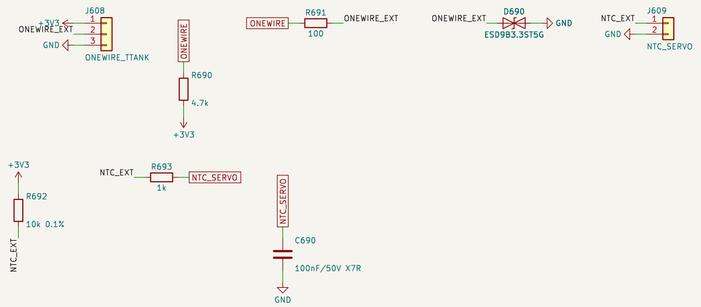
Thermocouple 2 (case aft): MAX31856, K type, 100 ohm + 100 nF + 10 nF filter (Maxim 19-7534 Fig. 8)



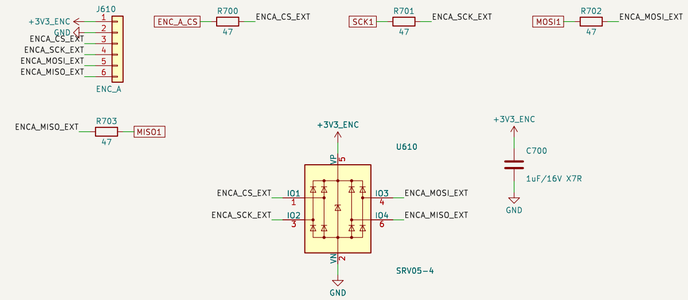
Thermocouple 3 (nozzle holder): MAX31856, K type, 100 ohm + 100 nF + 10 nF filter (Maxim 19-7534 Fig. 8)



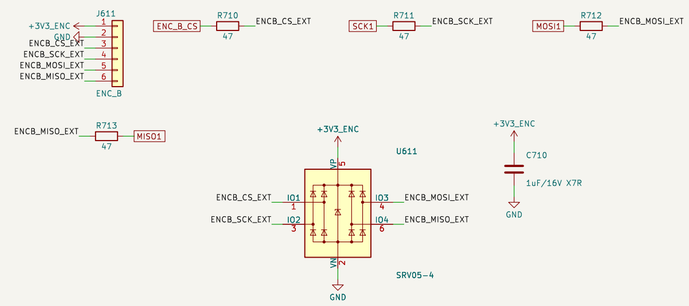
Tank wall 1-Wire (DS18B20) and pintle-servo bracket NTC



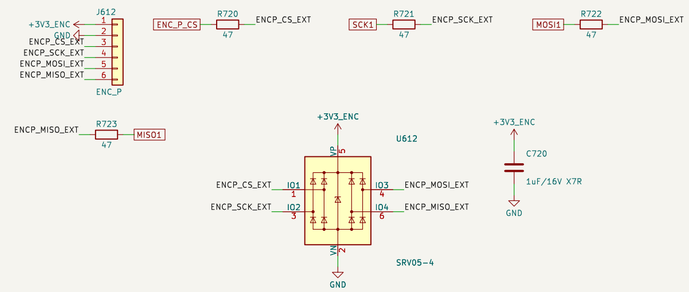
Encoder gimbal axis A: AS5048A (3.3 V mode) over shielded cable, 47 ohm series, ESD



Encoder gimbal axis B: AS5048A (3.3 V mode) over shielded cable, 47 ohm series, ESD



Encoder pintle crank: AS5048A (3.3 V mode) over shielded cable, 47 ohm series, ESD



Status: design schematic. ERC clean is NOT hardware validation. Not built, not tested.
KiCad used: 7.0.11 (Ubuntu 24.04 package 7.0.11+dfsg-1build4)
Generated by hardware/kicad_gen/make_ecu_kicad.py (2026-10-12); edit freely in KiCad
ARGOS H1 engine — 06-electronics
Sheet: /Sensors/
File: sensors.kicad_sch
Title: ARGOS ECU v2.1 — ADC, pressure, load cell, TC, encoders
Size: A2 | Date: 2026-10-12 | **Rev: v2.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 7/7